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Debiased recommendation via diffusion-guided implicit graph augmentation with long-tail awareness

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ABSTRACT

Implicit feedback is widely adopted in recommender systems because it is easy to collect and provides broad behavioral coverage. However, observed implicit interactions may contain noise, suffer from popularity bias, and exhibit high sparsity, whereas unobserved user–item pairs cannot be directly treated as negative feedback. These issues constrain the model's ability to learn reliable user preferences and provide effective recommendations for long-tail items. Therefore, this study proposes a latent Preference Diffusion-based interaction Completion Graph Recommendation model, termed PDCGR. Specifically, we first design a graph-guided latent preference recovery process that exploits graph structural information to guide latent-space preference diffusion, thereby improving the robustness of user preference representations under sparse implicit feedback. Second, we develop a latent preference diffusion objective integrated with spectral regularization to prevent the recovered representations from becoming excessively concentrated along a few dominant singular directions, thereby reducing the influence of popularity-dominated patterns. Furthermore, we construct a diffusion-guided interaction completion mechanism that selects high-confidence candidate interactions from strictly unobserved user–item pairs and provides additional relations for subsequent graph propagation and recommendation learning. Experimental results on three public datasets demonstrate that PDCGR outperforms representative graph-based and diffusion-based recommendation methods. On average across all datasets, compared with the strongest baseline, PDCGR improves HR and NDCG by 6.69% and 6.50%, respectively. In addition, we also conduct extensive experiments to verify that PDCGR can reduce dependence on popular items, balance item exposure, and improve the ranking quality of long-tail items.

1. Introduction

Recommender systems have become essential for helping users efficiently access relevant contents in e-commerce, online video, and social media scenarios (Cao et al., 2025; Li et al., 2025; Zhou et al., 2025a). Most recommendation models rely on implicit feedback, such as clicks, browsing records, and purchases, due to its availability and low cost (Hu et al., 2008). However, implicit feedback provides an incomplete and potentially biased representation of user preferences: observed interactions are confounded by platform exposure mechanisms and item popularity rather than reflecting genuine user interest alone; whereas unobserved interactions may result from limited exposure rather than negative preferences (Li et al., 2026c; Wang et al., 2021). This exposure bias, coupled with the inherent sparsity of implicit feedback, induces a severe structural inequality in user-item interaction graph. As platform exposure mechanisms tend to popular items, a small number of head items accumulate a disproportionate share of observed interaction, resulting in most long-tail items with sparse and insufficient interaction. Recommendation models trained on such sparse and skewed observations often overly focus on reflecting popularity-dominated patterns (He et al., 2025a; Kang et al., 2026; Zhang et al., 2026), reinforcing head-item dominance and inadequately capturing users' potential preferences for long-tail items. Therefore, recovering reliable preference-relevant signals for long-tail items from sparse and exposure-biased implicit feedback, while preserving overall recommendation utility, remains a fundamental challenge in mitigating popularity bias.

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Early collaborative filtering(CF) methods infer user preferences through latent factorization (Koren et al., 2009), neural interaction modeling (He et al., 2017), or denoising autoencoders (Wu et al., 2016). However, these methods primarily model direct user-item associations and largely overlook the high-order collaborative dependencies encoded in interaction graphs. Graph-based recommender systems address this by representing interactions as user-item bipartite graphs and performing neighborhood aggregation and multi-hop propagation, as exemplified by NGCF (Wang et al., 2019) and its simplified variant LightGCN (He et al., 2020). Recent studies have further enriched graph representations through self-supervised learning (Deng et al., 2024b; Hao et al., 2025a; Li et al., 2025; Zhou et al., 2024), relation-aware propagation (Tian et al., 2026), intent-aware modeling (Tang et al., 2026), fine-grained behavioral pattern awareness (Li et al., 2026b), and multi-task learning of sequential behaviors (Wang et al., 2026). Despite their effectiveness in capturing collaborative signals, conventional topology-driven propagation methods are inherently vulnerable to the skewness of the observed interaction graphs, which may not fully exploit the semantic and supervisory information underlying user-item interactions. Because repeated graph propagation over a popularity-skewed structure inevitably amplifies the connectivity advantage of head items, yielding representations that are dominated by popularity rather than genuine user preferences. Hence, mitigating popularity bias requires improving the reliability of preference-relevant connectivity of long-tail items, rather than merely propagating information over the original popularity-skewed graph.

Existing debiasing methods reduce head-item dominance from implicit feedback through noise identification (Deng et al., 2024a; Zeng et al., 2025), hard-sample discrimination (Song et al., 2026), informative negative sampling (Deng et al., 2026; Zhang et al., 2025b), popularity-aware regularization (Lin et al., 2025), and debiased graph aggregation (Elahi et al., 2025). Despite their effectiveness, these methods primarily intervene at the sample, objective, or aggregation level while retaining the original popularity-skewed interaction graph. Long-tail items therefore remain represented by sparse neighborhoods, and potentially relevant but unobserved connections are unavailable for collaborative propagation. Graph augmentation provides a structural alternative by modifying observed connectivity or constructing auxiliary graph views (He et al., 2026). Existing strategies are categorized into random perturbations via edge dropout or addition (Sun et al., 2024), heuristic augmentation based on similarity or popularity (Zhou et al., 2025b), learnable augmentation via edge predictors or reinforcement learning (Li et al., 2026a; Tang et al., 2026). However, commonly used random or heuristic perturbations cannot ensure that the introduced edges align with users' latent preferences, and the recovery strength is not adaptively adjusted across items of different popularity levels. These limitations motivate an implicit graph augmentation mechanism, which can recover reliable preference-relevant connections in a long-tail-aware manner, adaptively adjusting the recovery across item-popularity levels.

Diffusion models have recently been introduced to recommendation as an iterative generative paradigm, consisting of forward perturbation and reverse denoising (Deng et al., 2024a; Wang et al., 2024). Existing diffusion-based methods cover various applications including preference reconstruction (Wang et al., 2023; Wen et al., 2026; Xia et al., 2025), interaction denoising (He et al., 2026; Li et al., 2024), representation refinement (Hao et al., 2025b; Yuan et al., 2025; Zhao et al., 2024), and graph structure enhancement (Cui et al., 2025; Zhang et al., 2025c). By progressively reconstructing user interaction profiles, the reverse denoising process can generate candidate user-item connections absent from the observed interaction graphs, providing a learnable alternative to random or heuristic graph augmentation. Subsequently, hierarchy-aware diffusion has further been combined with knowledge-enhanced contrastive learning to generate task-adaptive graph views and filter semantically irrelevant generated samples in hyperbolic space (Li et al., 2026a). Despite these advances, a fundamental limitation remains: diffusion models are typically optimized to reconstruct the empirical interaction distribution, so the reverse denoising process tends to assign disproportionately high recovery probabilities to frequently observed head items (Yang et al., 2025; Zhao et al., 2026; Zheng et al., 2026). While recent studies have explored popularity-aware conditional guidance (He et al., 2025b) and debiased representation enhancement (Zhang et al., 2025a), explicit control of popularity effects during iterative interaction generation—particularly with spectral regularization tailored for long-tail protection—still require in-depth exploration. Therefore, incorporating spectral regularization into reverse diffusion to suppress popularity-driven recovery while preserving reliable preference signals for long-tail items remains an important challenge for diffusion-guided graph augmentation.

To address the above challenges, this study proposes a latent Preference Diffusion-based interaction Completion Graph Recommendation model(PDCGR) to complete user-item interaction and mitigate popularity bias. First, we employ LightGCN to encode the observed user-item interaction graph, capturing high-order collaborative signals and generating initial user and item representations. Second, these graph-enhanced embeddings serve as conditional priors that guide an iterative diffusion process, wherein latent user preferences are recovered via reverse denoising. Subsequently, a spectral regularizer is incorporated into the reverse denoising process to constrain the recovery dynamics and popularity-dominated directions. Finally, we introduce a diffusion-guided completion strategy that estimates matching scores for unobserved interactions and selectively augments the original graph with high-confidence candidates, enabling robust neighborhood propagation for recommendation. The main contributions of this study are summarized as follows:

1. A graph-guided latent preference generation framework is proposed, integrating graph-based representation learning with diffusion-based preference modeling. It leverages graph propagation to capture high-order collaborative information and a denoising process to iteratively refine user representations, thereby achieving robust preference embeddings under sparse and noisy implicit feedback.
2. A spectral-regularized diffusion objective is developed to counteract popularity bias by constraining reverse denoising along popularity-dominated directions. This regularization, combined with a diffusion-guided interaction completion mechanism, enables the framework to transform recovered preferences into meaningful graph augmentation, thereby establishing a close-loop synergy between preference denoising and structure enrichment.
3. Extensive experiments on three public datasets demonstrate that PDCGR not only outperforms state-of-the-art baselines in recommendation accuracy, but also achieves more balanced item exposure and superior long-tail recommendation performance.

The remainder of this study is organized as follows: Section 2 introduces the preliminaries of diffusion models; Section 3 presents the proposed method PDCGR; Section 4 reports and analyzes the experimental results; Section 5 discusses the experimental findings and limitations. Finally, Section 6 concludes this work.

2. Preliminaries

A diffusion model is a likelihood-based generative model (Nichol & Dhariwal, 2021). Its core idea is to gradually inject Gaussian noise into data during the forward process so that the data distribution progressively approaches a known distribution, and then to learn a parameterized denoising transition in the reverse process to gradually recover the target from noisy states. The core principle is described as follows.

2.1. Forward diffusion

Let $\mathbf{z}_t$ denote the latent vector after t diffusion steps, where $0 \le t \le T$. At the initial step $t = 0$, $\mathbf{z}_0$ represents the clean latent representation

of the sample; at the final step $t = T$, $\mathbf{z}_T$ gradually approaches a pure Gaussian noise vector. In the forward diffusion process, Gaussian noise is progressively injected into $\mathbf{z}_{t-1}$ to obtain $\mathbf{z}_t$, such that the data distribution gradually evolves from the original latent distribution toward an approximate standard Gaussian distribution. For tractable modeling, this process is usually formulated as a Markov chain, which is shown as follows:

$$q(\mathbf{z}_t \mid \mathbf{z}_{t-1}) = \mathcal{N}\Big(\mathbf{z}_t; \sqrt{1-\beta_t}\,\mathbf{z}_{t-1}, \beta_t \mathbf{I}_d\Big), \quad 1 \le t \le T, \tag{1}$$

where $\beta_t \in (0,1)$ is the noise coefficient at step t, which controls the intensity of the diffusion process, and $\mathbf{I}_d$ is the d-dimensional identity matrix.

By exploiting the additive property of Gaussian distributions, the multi-step forward diffusion process can be expanded along the time dimension, yielding an equivalent form that depends only on the initial state $\mathbf{z}_0$ and the noise coefficients β_t. Given an independent Gaussian noise $\boldsymbol{\epsilon}$, for any time step t, we have

$$\mathbf{z}_t = \sqrt{\bar{\alpha}_t}\mathbf{z}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \tag{2}$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}_d)$ and $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$.

2.2. Reverse denoising

Corresponding to the forward diffusion process $q(\mathbf{z}_t \mid \mathbf{z}_{t-1})$, the reverse denoising process aims to gradually recover the least noisy latent vector $\mathbf{z}_{t-1}$ from the noisy vector $\mathbf{z}_t$ and finally return to the diffusion starting point $\mathbf{z}_0$. To this end, a learnable parameterized reverse transition distribution is adopted to characterize the denoising process, which is formulated as a Gaussian distribution, namely:

$$p_\theta(\mathbf{z}_{t-1} \mid \mathbf{z}_t) = \mathcal{N}\big(\mathbf{z}_{t-1}; \mu_\theta(\mathbf{z}_t, t), \tilde{\beta}_t \mathbf{I}_d\big), \quad t \in \{1, \dots, T\}, \tag{3}$$

$$\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t, \tag{4}$$

where $\tilde{\beta}_t$ denotes the effective variance computed from the forward diffusion parameters β_t and $\bar{\alpha}_t$, and $\mu_\theta(\mathbf{z}_t, t)$ is the denoising mean function.

To model $\mu_\theta(\mathbf{z}_t, t)$, a time-conditioned denoising network (SDNet) composed of two fully connected layers is adopted. For the current time step t, the corresponding latent vector $\mathbf{z}_t$ and the time embedding vector $\boldsymbol{h}_t$ are concatenated, which is fed into the network. This process can be formulated as follows:

$$\mu_\theta(\mathbf{z}_t, t) = \mathrm{FC}^2\big([\mathbf{z}_t \,\|\, \boldsymbol{h}_t]\big), \tag{5}$$

$$\mathrm{FC}(\boldsymbol{x}) = \delta(\mathbf{W}\boldsymbol{x} + \boldsymbol{b}), \tag{6}$$

where $\|$ and $\delta(\cdot)$ denote vector concatenation and the nonlinear activation function, respectively. By iteratively sampling from $t = T$ to $t = 1$, the denoised preference representation can be progressively recovered, yielding the final reconstructed result.

To maximize the log-likelihood of the initial latent vector $\mathbf{z}_0$, i.e., $\log p_\theta(\mathbf{z}_0)$, variational inference is adopted to maximize its evidence lower bound (ELBO). According to the transition distribution between adjacent steps in the forward diffusion process, the joint distribution of the forward process can be expressed as

$$q(\mathbf{z}_{1:T} \mid \mathbf{z}_0) = \prod_{t=1}^{T} q(\mathbf{z}_t \mid \mathbf{z}_{t-1}), \tag{7}$$

the lower bound of $\log p_\theta(\mathbf{z}_0)$ can be derived as

$$\begin{aligned} &\log p_\theta(\mathbf{z}_0) \ge \mathcal{L}_{\mathrm{ELBO}} \\ &= \underbrace{\mathbb{E}_{q(\mathbf{z}_{1:T}|\mathbf{z}_0)}\big[\log p_\theta(\mathbf{z}_0 \mid \mathbf{z}_1)\big]}_{\text{reconstruction term}} - \underbrace{D_{\mathrm{KL}}\big(q(\mathbf{z}_T \mid \mathbf{z}_0) \,\|\, p(\mathbf{z}_T)\big)}_{\text{prior matching term}} \\ &\quad - \sum_{t=2}^{T} \underbrace{\mathbb{E}_{q(\mathbf{z}_{1:T}|\mathbf{z}_0)}\big[D_{\mathrm{KL}}\big(q(\mathbf{z}_{t-1} \mid \mathbf{z}_t, \mathbf{z}_0) \,\|\, p_\theta(\mathbf{z}_{t-1} \mid \mathbf{z}_t)\big)\big]}_{\text{denoising matching term}}. \end{aligned} \tag{8}$$

The three terms in Eq. (8) correspond to the reconstruction term, the prior matching term, and the stepwise denoising matching term, respectively. Since the first two terms have relatively limited influence on the parameter optimization, they can be treated as constants. Therefore, the loss function $\mathcal{L}_{\mathrm{ELBO}}$ mainly focuses on the stepwise denoising matching term, namely $D_{\mathrm{KL}}\big(q(\mathbf{z}_{t-1} \mid \mathbf{z}_t, \mathbf{z}_0) \,\|\, p_\theta(\mathbf{z}_{t-1} \mid \mathbf{z}_t)\big)$. Because both q and p_θ at time step t follow Gaussian distributions, the KL divergence can be transformed into the estimation error between the predicted clean latent vector $\hat{\mathbf{z}}_0^{(t)}$ and the initial latent vector $\mathbf{z}_0$. Following Ho et al. (2020), the denoising objective can be further simplified as

$$\mathcal{L}_{\mathrm{den}} = \mathbb{E}_{\mathbf{z}_0, t}\left[\left\|\mathbf{z}_0 - \hat{\mathbf{z}}_0^{(t)}\right\|_2^2\right]. \tag{9}$$

3. The PDCGR method

3.1. Overall framework

Under sparse implicit feedback scenarios, the effectiveness of recommendation often suffers from insufficient collaborative information, popular-interaction dominance, and the inability to recover user preference in the diffusion process. To address these issues, this study proposes a latent Preference Diffusion-based interaction Completion Graph Recommendation model (PDCGR). The proposed PDCGR model consists of four components: a user latent-space representation learning module, a spectral-regularized user preference diffusion module, an interaction completion graph model, and a prediction and optimization module. The overall framework is shown in Fig. 1. Specifically, PDCGR includes the following four components:

(1) Graph-based collaborative representation learning. This study constructs a user–item bipartite graph from observed implicit feedback(e.g., clicks) and employs a graph propagation mechanism to capture multi-hop collaborative dependencies. The learned user and item embeddings encode high-order structural signals and provide conditional priors for subsequent preference.

(2) Spectral-regularized latent preference recovery. Conditioned on the graph-derived user representations, the latent preference diffusion module progressively reconstructs user representations through reverse denoising. A spectral regularizer constrains popularity-dominated directions to reduce the influence of head-item patterns and preserve preference-relevant information.

(3) Diffusion-guided interaction completion and graph augmentation. Matching scores for unobserved user–item pairs are computed from the recovered user and item representations. High-confidence interactions are selected to augment the original graph, providing additional structural signals for recommendation.

(4) Prediction and optimization. Graph propagation and ranking optimization are performed on the augmented graph to strengthen long-tail collaborative signals, improve recommendation accuracy and mitigate popularity bias.

3.2. User preference representation module for implicit feedback

This module proposes a graph-guided latent preference generation approach (GLPG). Specifically, a lightweight graph convolutional network, namely LightGCN, is adopted as the graph encoder. It learns low-dimensional embeddings of the user set $\mathcal{U} = \{u_1, u_2, \dots, u_m\}$ and the item set $\mathcal{I} = \{i_1, i_2, \dots, i_n\}$, and aggregates global collaborative information to obtain reliable latent representations through the graph propagation mechanism. The details are described as follows.

In implicit feedback recommendation, the observed user–item interactions can be denoted by the interaction matrix $\mathbf{R} = (r_{ui})_{m\times n}$, where $r_{ui} = 1$ indicates that user u has interacted with item i, and $r_{ui} = 0$ otherwise. Accordingly, the corresponding user–item bipartite graph is defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where the vertex set is $\mathcal{V} = \mathcal{U} \cup \mathcal{I}$ and the edge set is $\mathcal{E} = \{(u, i) \mid r_{ui} = 1,\ u \in \mathcal{U},\ i \in \mathcal{I}\}$. For subsequent modeling, the adja-

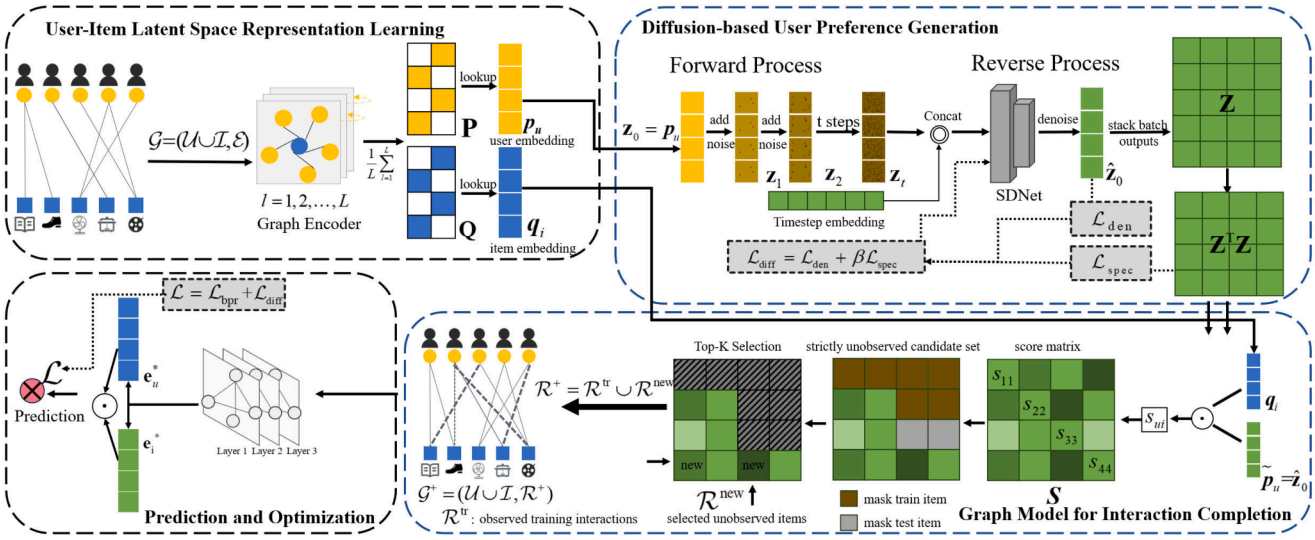


Fig. 1. Overall framework of the PDCGR model.

cency matrix of $\mathcal{G}$ is denoted by $\mathbf{A} \in \mathbb{R}^{(m+n)\times(m+n)}$, which is defined as

$$\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{R} \\ \mathbf{R}^\top & \mathbf{0} \end{bmatrix}. \tag{10}$$

To avoid the scale bias caused by nodes with different degree numbers during graph propagation, we construct the corresponding degree matrix $\mathbf{D} \in \mathbb{R}^{(m+n)\times(m+n)}$ based on the degree number of each vertex $v \in \mathcal{V}$. On this basis, the input matrix $\mathbf{A}$ of the graph propagation mechanism is symmetrically normalized, and the normalized matrix is given by

$$\tilde{\mathbf{A}} = \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}. \tag{11}$$

At the graph encoding stage, we first initialize the learnable embedding matrix of all nodes as

$$\mathbf{E}^{(0)} = \begin{bmatrix} \mathbf{P}^{(0)} \\ \mathbf{Q}^{(0)} \end{bmatrix} \in \mathbb{R}^{(m+n)\times d}, \tag{12}$$

where $\mathbf{P}^{(0)} \in \mathbb{R}^{m\times d}$ and $\mathbf{Q}^{(0)} \in \mathbb{R}^{n\times d}$ denote the initial user and item embedding matrices, respectively, and d is the latent dimensionality.

Taking $\mathbf{E}^{(0)}$ as an input, LightGCN performs multi-layer propagation to iteratively aggregate neighborhood information of graph nodes, updating the node representations on $\mathcal{G}$. The propagation rule at the lth layer is defined as

$$\mathbf{E}^{(l)} = \tilde{\mathbf{A}}\mathbf{E}^{(l-1)}, \quad l = 1, 2, \ldots, L. \tag{13}$$

Then, the embeddings from all layers, including the initial embeddings, are averaged to obtain the final node representations learned by the graph encoder, i.e.,

$$\mathbf{E} = \frac{1}{L+1}\sum_{l=0}^{L}\mathbf{E}^{(l)} = \begin{bmatrix} \mathbf{P} \\ \mathbf{Q} \end{bmatrix} \in \mathbb{R}^{(m+n)\times d}, \tag{14}$$

where $\mathbf{P} \in \mathbb{R}^{m\times d}$ and $\mathbf{Q} \in \mathbb{R}^{n\times d}$ denote the enhanced user and item embeddings in the latent space, respectively.

3.3. Spectral-regularized user preference diffusion module

In implicit feedback recommendation, although graph propagation can aggregate user neighborhood information, the learned representation of user u, denoted by $\mathbf{p}_u = \mathbf{P}_{u,:} \in \mathbb{R}^d$, is still constrained by limited observed interactions and thus cannot fully capture the user's latent interests. To address this issue, this study constructs a spectral-regularized diffusion model in the user latent space, termed the SpecReg-based Diffusion Model (SpeDM). SpeDM can generate complete and high-quality user preference representations by performing the forward noising and reverse denoising processes. The details are described as follows.

In the forward noising process, we set the initial state as $\mathbf{z}_0 = \mathbf{p}_u$ and progressively inject Gaussian noise along the time dimension to obtain the corresponding noisy latent vector $\mathbf{z}_t$ $(1 \le t \le T)$. Next, SDNet with two-layer fully connected denoising network is constructed, modeling the reverse denoising process. By gradually recovering $\mathbf{z}_t$, the model predicts the user preference representation as $\hat{\mathbf{z}}_0(t) = \mu_\theta(\mathbf{z}_t, t)$, where the parameters θ in the denoising network $\mu_\theta(\cdot)$ are shared across all time steps t $(1 \le t \le T)$. For convenience, after the diffusion model converges, we denote the denoised output of user u as the latent preference vector $\tilde{\mathbf{p}}_u = \hat{\mathbf{z}}_0 \in \mathbb{R}^d$. Then, the denoised user preference matrix of all users is written as $\tilde{\mathbf{P}} = [\tilde{\mathbf{p}}_1; \ldots; \tilde{\mathbf{p}}_m] \in \mathbb{R}^{m\times d}$. Combining $\tilde{\mathbf{P}}$ with the item embedding matrix $\mathbf{Q} = [\mathbf{q}_1; \ldots; \mathbf{q}_n] \in \mathbb{R}^{n\times d}$ learned by the graph encoder, we can compute the candidate interaction confidence matrix $\mathbf{S} = (s_{ui})_{m\times n}$, measuring the reliability of different user–item pairs as potential completion edges. $\mathbf{S}$ is defined as

$$\mathbf{S} = \tilde{\mathbf{P}}\mathbf{Q}^\top \in \mathbb{R}^{m\times n}, \tag{15}$$

where $s_{ui} = \tilde{\mathbf{p}}_u^\top \mathbf{q}_i$ denotes the matching score between the denoised preference of user u and the embedding of item i.

However, when the diffusion model is trained only with the conventional denoising loss $\mathcal{L}_{\text{den}}$, the recovered user representations may remain concentrated along a small number of dominant directions associated with frequently observed interaction patterns. Motivated by the spectral analysis of popularity amplification in ReSN (Lin et al., 2025), we introduce a spectral regularizer based on the Frobenius norm of the Gram matrix of the recovered user representations to constrain these dominant directions. Specifically, let the reverse outputs in one batch be stacked row by row into the matrix $\mathbf{Z} = [\hat{\mathbf{z}}_{0,1}, \hat{\mathbf{z}}_{0,2}, \ldots, \hat{\mathbf{z}}_{0,B}]^\top \in \mathbb{R}^{B\times d}$, where B is the number of users in one batch. The corresponding spectral regularization loss is defined as

$$\mathcal{L}_{\text{spec}} = \left\|\mathbf{Z}^\top\mathbf{Z}\right\|_F^2. \tag{16}$$

From the singular-value perspective, this regularization term imposes stronger penalties on dominant representation directions with large singular values, thereby alleviating excessive spectral concentration in the recovered user representations. Meanwhile, the denoising loss preserves preference reconstruction, allowing the two objectives to jointly balance preference recovery and spectral regularization. Accordingly, the overall loss of user preference diffusion in the latent space is described as

$$\mathcal{L}_{\text{diff}} = \mathcal{L}_{\text{den}} + \beta\mathcal{L}_{\text{spec}}, \tag{17}$$

where $\beta > 0$ is the hyperparameter controlling the strength of spectral regularization. Spectral constraints are continuously imposed at each time step of the reverse denoising process, thereby being jointly optimized with the denoising mechanism.

3.4. Recommendation prediction module based on interaction completion graph model

The proposed SpeDM module mainly focuses on the computation of matching scores between user and item representations, while overlooking the effective exploitation of high-order information embedded in the original interaction graph.

Implicit feedback recommendation can be viewed as a positive–unlabeled learning problem, since unobserved user–item pairs may contain potential positive interactions rather than being uniformly regarded as negative feedback (Zhou et al., 2021). Moreover, previous graph recommendation research has shown that representation-based Top-K sampling can augment the user–item interaction matrix and provide additional neighbors for subsequent graph propagation (Fan et al., 2023).

To better capture users' latent true preferences and avoid repeatedly reinforcing existing interactions, we further propose a diffusion-guided interaction completion strategy. Specifically, for each user, new candidate interaction items are selected from the strictly unobserved set, and the corresponding item candidate set is constructed as

$$C_u = \left\{ i \in \mathcal{I} \mid (u,i) \notin \mathcal{E}^{\mathrm{tr}},\ (u,i) \notin \mathcal{E}^{\mathrm{te}} \right\}, \tag{18}$$

where $\mathcal{E}^{\mathrm{tr}}$ and $\mathcal{E}^{\mathrm{te}}$ denote the training interaction set and the test interaction set, respectively.

Based on Eq. (15), the matching score s_{ui} between user u and item i is used to rank the items in C_u. Then, the Top-k_{new} items are selected to form the completion candidate set, i.e.,

$$\Omega_u^{\mathrm{new}} = \text{Top-}k_{\mathrm{new}}\left(\left\{ (i, s_{ui}) \mid i \in C_u \right\} \right), \tag{19}$$

where k_{new} is a relatively small hyperparameter that controls the number of completed interactions for each user. This design helps capture users' latent interests while preventing too many newly added edges from distorting the original interaction structure.

Accordingly, the interaction completion set $\mathcal{E}^{\mathrm{new}} = \bigcup_{u \in \mathcal{U}} \left\{ (u,i) \mid (i, s_{ui}) \in \Omega_u^{\mathrm{new}} \right\}$ is generated by merging the newly selected interactions of all users. Then, the augmented interaction set is defined as $\mathcal{E}^{+} = \mathcal{E}^{\mathrm{tr}} \cup \mathcal{E}^{\mathrm{new}}$. On this basis, the corresponding augmented bipartite graph is denoted by $\mathcal{G}^{+} = (\mathcal{U} \cup \mathcal{I}, \mathcal{E}^{+})$. Then, the proposed GLPG method in Section 3.2 is applied to update the node representations on the augmented bipartite graph $\mathcal{G}^{+}$. Specifically, graph propagation is performed on the adjacency matrix of $\mathcal{G}^{+}$ and its symmetrically normalized form according to Eqs. (11)–(14). The resulting node embeddings on the augmented graph are denoted by $\mathbf{E}^{*}$.

Let $\mathbf{e}_u^{*}$ and $\mathbf{e}_i^{*}$ denote the row vectors in $\mathbf{E}^{*}$ corresponding to user u and item i, respectively. Rather than directly using the diffusion-based matching scores as final predictions, the completed interactions are incorporated into the graph structure, allowing the recovered preference signals to participate in subsequent neighborhood aggregation. Then, the predicted matching score between user u and item i is computed by the inner product as

$$\hat{y}_{ui} = \left\langle \mathbf{e}_u^{*}, \mathbf{e}_i^{*} \right\rangle, \tag{20}$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product operation.

To improve recommendation accuracy, the pairwise Bayesian Personalized Ranking (BPR) loss (Rendle et al., 2009) is adopted on the augmented interaction set $\mathcal{E}^{+}$. Specifically, for any positive interaction $(u,i) \in \mathcal{E}^{+}$, a negative item j is randomly sampled from the unobserved items of user u, satisfying $(u,j) \notin \mathcal{E}^{+}$. Then, the ranking loss corresponding to the triplet (u,i,j) is defined as

$$\mathcal{L}_{\mathrm{bpr}} = -\sum_{(u,i,j)} \log \sigma\left(\hat{y}_{ui} - \hat{y}_{uj} \right), \tag{21}$$

where $\sigma(\cdot)$ is the Sigmoid function.

Since the latent-space user preference diffusion module and the interaction completion graph model jointly optimize the recommendation performance, the final objective function of the proposed PDCGR model is defined as

$$\mathcal{L} = \mathcal{L}_{\mathrm{den}} + \beta \mathcal{L}_{\mathrm{spec}} + \mathcal{L}_{\mathrm{bpr}}. \tag{22}$$

In summary, the training procedure of the proposed PDCGR model is presented in Algorithm 1.

Algorithm 1 PDCGR training.

Input: user–item interaction graph $\mathcal{G}$, augmentation size k_{new}
Initialize model parameters Θ
repeat
  Compute user and item embeddings on $\mathcal{G}$ by Eqs. (11)–(14)
  Set $\mathbf{z}_0 = \mathbf{p}_u$ and perform diffusion denoising by Eqs. (1)–(5)
  Optimize the diffusion objective by Eqs. (17)
until converged
Construct the candidate set and select k_{new} items by Eq. (18) and Eq. (19)
Construct the augmented graph $\mathcal{G}^{+}$
repeat
  Perform graph propagation on $\mathcal{G}^{+}$ by Eqs. (11)–(14)
  Compute the prediction score and optimize BPR by Eq. (20) and Eq. (21)
until converged
Output: optimized Θ

3.5. Complexity analysis

Let L, h, and B be the number of graph propagation layers, the dimensionality of the time-step embedding in the diffusion process, and the number of users in each batch, respectively. The overall time complexity of PDCGR is $\mathcal{O}\left(\left(|\mathcal{E}| + |\mathcal{E}^{+}| \right) dL + (|\mathcal{U}| + |\mathcal{I}| + B)d^2 + Bdh + |\mathcal{U}||\mathcal{I}|d \right)$, where $|\mathcal{E}|$ and $|\mathcal{E}^{+}|$ denote the numbers of edges in the original graph and the augmented graph, respectively. Specifically, the time complexity of latent-space preference diffusion is approximately $\mathcal{O}(Bd(d+h))$, since the denoising network mainly involves transformations on the latent representation and the time-step embedding. The time complexity of graph propagation in the interaction completion graph prediction module is linear with respect to the number of graph edges and the embedding dimensionality, i.e., $\mathcal{O}\left(\left(|\mathcal{E}| + |\mathcal{E}^{+}| \right) dL \right)$. In addition, computing the user–item matching scores for interaction completion requires $\mathcal{O}(|\mathcal{U}||\mathcal{I}|d)$ time. The graph encoding stage also includes the transformation of user and item embeddings, whose complexity is $\mathcal{O}\left((|\mathcal{U}| + |\mathcal{I}|)d^2 \right)$.

Accordingly, the overall space complexity of PDCGR is $\mathcal{O}\left((|\mathcal{U}| + |\mathcal{I}|)d + |\mathcal{E}| + |\mathcal{E}^{+}| + d^2 + dh \right)$. This complexity remains acceptable in practice. More specifically, it consists of four parts: storing the user and item embedding matrices, which requires $\mathcal{O}((|\mathcal{U}| + |\mathcal{I}|)d)$, storing the sparse adjacency structure of the original graph, which requires $\mathcal{O}(|\mathcal{E}|)$, storing the sparse adjacency structure of the augmented graph, which requires $\mathcal{O}(|\mathcal{E}^{+}|)$, and storing the parameters in the latent-space diffusion module, whose complexity is $\mathcal{O}(dh + d^2)$.

4. Experiments

In this section, we conduct extensive experiments to evaluate the effectiveness of the proposed PDCGR model and answer the following research questions.

RQ1: How does our model perform compared with other methods?
RQ2: Does our model successfully mitigate popularity bias?
RQ3: What is the optimal backbone architecture of our model?

Table 1
Description of dataset information.

Dataset	#Users	#Items	#Interactions	#Sparsity
ML-100K	943	1682	100,000	93.7%
ML-1M	6040	3952	1,000,209	95.8%
AIV	5130	1685	37,126	99.6%

RQ4: How do different interaction completion strategies affect recommendation performance?
RQ5: How does each component affect our model?
RQ6: How sensitive is our model to different parameter settings?

4.1. Experimental settings

This section describes the datasets used in the experiment, evaluation metrics, benchmark models, and the experimental implementation details.

4.1.1. Datasets

To comprehensively evaluate the recommendation performance of the proposed PDCGR model under different data scales and sparsity levels, we conduct experiments on three public datasets, namely MovieLens 100K (ML-100K), MovieLens 1M (ML-1M), and Amazon Instant Video (AIV). The MovieLens datasets were collected by the GroupLens research group from a movie recommendation platform and contain user feedback on movies. The AIV dataset is a publicly available benchmark collected from the Amazon e-commerce platform and contains user feedback on instant video content. The statistical information of the three datasets is summarized in Table 1.

Following the leave-one-out protocol, each dataset is split into a training set and a test set. Specifically, for each user, the most recent interaction is treated as the positive test instance, while the remaining interactions are used for training. During the testing stage, 99 items that the user has not interacted with are randomly sampled as negative instances and combined with the positive test item to form the candidate set. The model is required to predict preference scores for all items in the candidate set and rank them accordingly. After training, the Top-K items with the highest predicted scores are selected as the final recommendation list.

4.1.2. Evaluation metrics

To comprehensively evaluate the recommendation accuracy of the proposed PDCGR model, we adopt two widely used metrics in recommender systems, namely Hit Ratio (HR) and Normalized Discounted Cumulative Gain (NDCG). HR measures whether the ground-truth item appears in the recommended list, while NDCG further considers the ranking position of the hit item. A higher value of HR@K or NDCG@K indicates better recommendation performance.

For the Top-K recommendation list, the two metrics are defined as

$$\mathrm{HR@K} = \frac{1}{|\mathcal{U}|}\sum_{u\in\mathcal{U}} \mathrm{hit}(L_{u,\mathrm{K}}), \tag{23}$$

$$\mathrm{NDCG@K} = \frac{1}{|\mathcal{U}|}\sum_{u\in\mathcal{U}} \frac{1}{\log_2(\mathrm{rank}_u + 1)}, \tag{24}$$

where $\mathrm{hit}(L_{u,\mathrm{K}})$ indicates whether the positive test item of user u appears in the Top-K recommendation list. Specifically, $\mathrm{hit}(L_{u,\mathrm{K}}) = 1$ if the positive test item is included in the list, and $\mathrm{hit}(L_{u,\mathrm{K}}) = 0$, otherwise. Here, rank_u denotes the ranking position of the positive test item in the recommendation list of user u. If the positive test item does not appear in the Top-K list, its contribution to NDCG@K is set to 0. Therefore, larger values of HR@K and NDCG@K indicate better performance on the corresponding Top-K recommendation task.

Table 2
Hyperparameter settings of PDCGR.

Hyperparameter	ML-100K	ML-1M	AIV
Learning rate η	10^{-3}	10^{-3}	10^{-3}
DNN embedding dimension d	64	128	128
Diffusion steps T	100	200	200
Time-step embedding	8	16	16
Noise scale β_s	10^{-1}	10^{-1}	10^{-1}
Minimum noise value $\sigma_{\min}$	10^{-4}	10^{-3}	10^{-3}
Maximum noise value $\sigma_{\max}$	10^{-2}	10^{-2}	10^{-2}
Number of new interactions k_{new}	2	2	2
Batch size	1024	1024	4096
Spectral regularization weight β	0.01	0.01	0.001
Graph propagation layers L	3	3	3

Table 3
Configuration of the experimental environment.

Configuration	Details
Operating system	Windows 10
GPU	NVIDIA GeForce RTX 4060
GPU memory	8 GB
Python	3.8.20
PyTorch	2.1.0 + cu121

4.1.3. Hyperparameter settings

All methods are implemented in PyTorch, optimized with Adam, and executed five runs independently with different random seeds. To ensure fairness and reproducibility, each model is tuned under its best hyperparameter configuration. For PDCGR, the learning rate is selected from $\eta \in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$, the embedding dimension from $d \in \{32, 64, 128\}$, the diffusion steps from $T \in \{50, 100, 200, 300\}$, and the graph propagation layers from $L \in \{1, 2, 3, 4\}$. The spectral regularization weight and the number of newly completed interactions are selected from $\beta \in \{0, 0.001, 0.01, 0.1, 0.5\}$ and $k_{\mathrm{new}} \in \{0, 1, 2, 3, 5, 10\}$, respectively. The final hyperparameter settings of the proposed PDCGR model on the three datasets are reported in Table 2.

4.1.4. Environment configuration

All experiments are conducted under the same hardware and software environment to ensure a fair runtime comparison. The detailed environment configuration is summarized in Table 3.

4.2. Comparative performance analysis

4.2.1. Baselines

To verify the effectiveness of the proposed PDCGR model and ensure the fairness of comparative experiments, we select a diverse set of representative baselines, covering traditional collaborative filtering, graph-based recommendation, and diffusion-based recommendation methods, as well as a spectral debiasing method. The compared baselines are described as follows.

1. **MF** (Koren et al., 2009): A classical matrix factorization method decomposes the user–item interaction matrix into low-dimensional user and item latent vectors, and models user preferences through vector inner products.
2. **NCF** (He et al., 2017): A neural collaborative filtering model combines embedding layers and multilayer perceptrons to capture nonlinear interactions between users and items.
3. **ICDAE** (Liu et al., 2023): An improved collaborative denoising autoencoder model takes noisy user interaction vectors as the input and learns latent user representations through reconstruction, alleviating noise interference in implicit feedback.
4. **NGCF** (Wang et al., 2019): A graph convolutional collaborative filtering method performs multi-layer neighborhood aggregation on the user–item bipartite graph, propagating high-order collaborative information into user and item embeddings.

5. **LightGCN** (He et al., 2020): A simplified graph collaborative filtering model removes feature transformation and nonlinear activation from conventional graph convolutional networks, while preserving linear neighborhood propagation and layer-wise aggregation to efficiently exploit high-order graph structure information.
6. **DiffRec** (Wang et al., 2023): diffusion-based recommendation model that applies forward noising and reverse denoising to user interaction vectors to reconstruct user preferences and generate recommendation lists.
7. **RecDiff** (Li et al., 2024): A diffusion-based denoising framework that performs multi-step noise injection and removal in a compressed hidden space to reduce the influence of unreliable relations on user representations.
8. **ReSN** (Lin et al., 2025): A recommendation model introduces spectral norm regularization into the matrix factorization framework. By constraining the dominant singular values of the score matrix, it alleviates popularity bias caused by the dominance of popular items in the latent space.
9. **GDiffMAE** (Zhang et al., 2025c): A graph recommendation model that integrates an adaptive mask encoder for structural reconstruction with a guided diffusion model for semantic reconstruction.
10. **ConDiff** (Wen et al., 2026): A conditional graph diffusion model that incorporates personalized collaborative signals and user–item interaction information as conditions for reverse generation to reconstruct user–item interactions.

4.2.2. Overall performance analysis (RQ1)

This study conducts comparative experiments on the three public datasets mentioned above. The overall performance of the proposed PDCGR model compared to baseline models is shown in Table 4. Moreover, to verify the statistical significance of the performance gains of PDCGR over comparison methods, we conduct a paired t-test with a significance level of $\alpha = 0.05$ on the NDCG@10 metric. The results from Table 4 indicate that all improvements over baselines are statistically significant, demonstrating the effectiveness of our model in recommendation scenarios.

Analyzing applicability across different datasets. It is clear that PDCGR achieves the best results on all evaluation metrics across the three datasets. On the relatively dense ML-100K dataset, PDCGR achieves 10.21% and 8.75% relative improvements in HR@3 and NDCG@3, respectively, over the best-performing baselines, RecDiff and GDiffMAE. On the larger ML-1M dataset, PDCGR achieves 5.51% and 7.00% relative improvements in HR@10 and NDCG@10, respectively, over the best-performing baselines, GDiffMAE and ConDiff. On the highly sparse AIV dataset, the advantage of PDCGR becomes more significant. Specifically, PDCGR improves HR@5 by 10.10% over GDiffMAE and NDCG@5 by 8.87% over ConDiff. These results demonstrate that PDCGR consistently outperforms the baseline models under different data scales and sparsity levels.

Exploring the high-order interaction modeling ability of the interaction completion graph model. As shown in Table 4, the collaborative filtering methods, including MF and NCF, achieve relatively poor performance in terms of HR@K and NDCG@K. This is because they rely mainly on matrix factorization or nonlinear mapping to learn user preferences, and thus cannot effectively capture the high-order collaborative relationships between users and items. To better model such relationships, NGCF and LightGCN introduce graph propagation and exploit multi-layer neighborhood aggregation to enhance user preference representations. According to Table 4, NGCF achieves better HR@10 and NDCG@10 on the sparse AIV dataset than NCF, while LightGCN consistently outperforms NGCF across datasets with different scales and sparsity levels. However, these graph-based baselines still lack an explicit mechanism for handling the noise in implicit feedback, and thus cannot sufficiently recover users' latent interests that have not been explicitly exposed. Compared with LightGCN, PDCGR further exploits an interaction completion graph model to mine implicit interaction relationships, which shows clear superiority on all HR@K and NDCG@K metrics across the three datasets with different scales and sparsity levels. This advantage remains notable even on the top-10 recommendation list, where the relative improvement is the smallest. Specifically, on the ML-100K, ML-1M, and AIV datasets, PDCGR improves HR@10 by 11.57%, 9.10%, and 10.36%, respectively, and improves NDCG@10 by 15.09%, 11.87%, and 11.91%, respectively. These results verify that the graph-guided user preference generation module in Section 3.2 and the interaction completion graph model in Section 3.4 can better exploit high-order interaction relationships through graph propagation, thereby enhancing the model's ability to represent users' latent preferences.

Evaluating denoising ability. Table 4 shows that DiffRec, which performs multi-step iterative denoising generation, achieves clearly better recommendation performance than ICDAE, which relies on single-step reconstruction denoising. RecDiff, which filters noisy input signals, further outperforms DiffRec. Compared with RecDiff, the proposed PDCGR model shows significant advantages on all HR@K and NDCG@K metrics across the three datasets with different scales and sparsity levels. In addition, the graph structure-enhanced models, including GDiffMAE and ConDiff, achieve relatively strong recommendation performance on the sparse AIV dataset. Compared with the best graph-enhanced baselines on the sparse AIV dataset, PDCGR achieves improvements of 6.74% in HR@3, 10.09% in HR@5, and 8.00% in HR@10, as well as 5.85% in NDCG@3, 8.87% in NDCG@5, and 6.86% in NDCG@10. Moreover, compared with ReSN, which introduces spectral regularization into the MF framework, the proposed PDCGR model improves HR@10 by 5.96%, 6.08%, and 8.69% on the three datasets, respectively, and improves NDCG@10 by 9.73%, 17.13%, and 8.53%, respectively. These results demonstrate that PDCGR effectively incorporates spectral constraints into the overall representation learning framework and achieves better coordination among graph structure modeling, latent-space diffusion modeling, and recommendation prediction. This coordination improves the quality of recovered preference representations and enhances the model's robustness to noise in implicit feedback.

In summary, the proposed PDCGR model achieves robust recommendation performance, denoising ability, and debiasing capability across three datasets with different scales and sparsity levels by exploiting high-order relationships in the interaction completion graph and incorporating spectral constraints. Its advantages are especially evident in sparse data scenarios.

4.2.3. Debiasing performance analysis (RQ2)

Due to the long-tail distribution of user–item interactions (Park & Tuzhilin, 2008; Zhang et al., 2025a), recommender models are prone to popularity bias, causing recommendation results to concentrate on head items. To comprehensively evaluate the debiasing ability of PDCGR, we analyze four complementary aspects: the dependence on historically popular items, the concentration of recommendation exposure, the proportion of long-tail items in recommendation lists, and the ranking quality for long-tail items. Accordingly, four metrics are adopted, including AvgPop@10, Gini@10, APLT@10, and Tail-NDCG@10. Lower AvgPop@10 and Gini@10 indicate weaker dependence on popular items and a more balanced item exposure distribution, respectively. Higher APLT@10 and Tail-NDCG@10 indicate a larger proportion of long-tail items and better ranking quality for long-tail ground-truth items, respectively.

AvgPop@10 used to measure the dependence of the recommendation list on popular items is defined as

$$\mathrm{AvgPop@K} = \frac{1}{|\mathcal{U}|}\sum_{u\in\mathcal{U}}\frac{1}{\mathrm{K}}\sum_{i\in\mathrm{R}_u^{\mathrm{K}}}\phi_i, \tag{25}$$

where $\mathrm{R}_u^{\mathrm{K}}$ denotes the Top-K recommendation list for user u, and ϕ_i denotes the historical popularity of item i, defined as its interaction frequency in the training set. To evaluate the concentration of item

Table 4
Evaluation results on three datasets.

Dataset	Methods	HR@3	NDCG@3	HR@5	NDCG@5	HR@10	NDCG@10	p-value	t
ML-100K	MF	0.348	0.272	0.442	0.313	0.623	0.369	3.24×10^{-5}	54
	NCF	0.334	0.258	0.454	0.318	0.631	0.377	3.81×10^{-5}	26
	ICDAE	0.305	0.247	0.417	0.293	0.604	0.355	2.60×10^{-8}	4
	NGCF	0.291	0.226	0.390	0.267	0.570	0.325	2.05×10^{-8}	32
	LightGCN	0.351	0.266	0.451	0.315	0.629	0.375	8.67×10^{-6}	15
	DiffRec	0.334	0.265	0.442	0.310	0.607	0.363	7.47×10^{-7}	76
	RecDiff	0.383	0.299	0.497	0.345	0.664	0.400	5.08×10^{-6}	165
	ReSN	0.371	0.285	0.496	0.336	0.663	0.394	1.80×10^{-5}	17
	GDiffMAE	0.382	0.304	0.489	0.358	0.664	0.410	4.80×10^{-4}	37
	ConDiff	0.345	0.277	0.440	0.317	0.636	0.386	8.11×10^{-6}	32
	PDCGR	**0.422**	**0.331**	**0.527**	**0.375**	**0.702**	**0.432**	–	29
ML-1M	MF	0.295	0.233	0.391	0.268	0.554	0.322	1.07×10^{-7}	658
	NCF	0.386	0.301	0.486	0.339	0.671	0.397	4.47×10^{-5}	527
	ICDAE	0.337	0.267	0.437	0.305	0.630	0.371	8.95×10^{-7}	91
	NGCF	0.379	0.296	0.488	0.337	0.657	0.395	9.24×10^{-8}	571
	LightGCN	0.389	0.315	0.504	0.356	0.670	0.414	2.13×10^{-4}	316
	DiffRec	0.366	0.290	0.468	0.330	0.623	0.382	4.64×10^{-6}	889
	RecDiff	0.419	0.327	0.528	0.368	0.680	0.413	1.39×10^{-6}	1194
	ReSN	0.428	0.338	0.536	0.378	0.689	0.395	2.52×10^{-6}	215
	GDiffMAE	0.426	0.336	0.533	0.377	0.693	0.431	7.01×10^{-5}	530
	ConDiff	0.433	0.346	0.536	0.386	0.674	0.433	3.62×10^{-6}	207
	PDCGR	**0.448**	**0.361**	**0.567**	**0.410**	**0.731**	**0.463**	–	339
AIV	MF	0.303	0.246	0.372	0.271	0.494	0.318	2.13×10^{-7}	267
	NCF	0.326	0.274	0.387	0.286	0.481	0.321	1.38×10^{-5}	79
	ICDAE	0.336	0.282	0.399	0.305	0.496	0.341	1.13×10^{-5}	85
	NGCF	0.302	0.242	0.389	0.275	0.549	0.329	4.66×10^{-6}	219
	LightGCN	0.365	0.291	0.458	0.328	0.580	0.379	1.35×10^{-5}	127
	DiffRec	0.359	0.303	0.428	0.315	0.518	0.361	1.22×10^{-5}	319
	RecDiff	0.377	0.303	0.465	0.337	0.601	0.387	1.26×10^{-4}	588
	ReSN	0.387	0.307	0.457	0.337	0.589	0.391	5.33×10^{-5}	175
	GDiffMAE	0.392	0.323	0.466	0.353	0.593	0.397	3.86×10^{-4}	122
	ConDiff	0.395	0.328	0.460	0.353	0.565	0.386	4.77×10^{-5}	166
	PDCGR	**0.421**	**0.348**	**0.513**	**0.384**	**0.640**	**0.424**	–	131

[1]The best performance is in boldface and the second-best performance is underlined.
[2]t denotes the average runtime (s) of one epoch.
[3]p-value is computed via a paired t-test against PDCGR (NDCG@10, two-tailed, $\alpha = 0.05$).

exposure, Gini@K is defined as

$$\text{Gini@K} = \frac{\sum_{i=1}^{|\mathcal{I}|} (2i - |\mathcal{I}| - 1) n_i^{\text{K}}}{|\mathcal{I}| \sum_{i=1}^{|\mathcal{I}|} n_i^{\text{K}}}, \tag{26}$$

where n_i^{K} denotes the ith smallest exposure frequency after all items are sorted in ascending order by the number of times they appear in users' Top-K recommendation lists. Unlike ϕ_i, which measures historical popularity, n_i^{K} measures recommendation exposure. To measure the proportion of long-tail items in recommendation lists, the Average Percentage of Long-Tail Items (APLT) is defined as

$$\text{APLT@K} = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \frac{\left| \text{R}_u^{\text{K}} \cap \mathcal{I}^{\text{tail}} \right|}{\text{K}}, \tag{27}$$

where $\mathcal{I}^{\text{tail}}$ contains the 80% least-popular items according to ϕ_i.

In this experiment, we select representative baseline models from traditional collaborative filtering, graph recommendation, and diffusion recommendation methods, including DiffRec (Wang et al., 2023), NCF (He et al., 2017), and LightGCN (He et al., 2020). As shown in Fig. 2, PDCGR achieves the lowest AvgPop@10 on all three datasets, demonstrating its lower dependence on historically popular items.

As shown in Fig. 3, PDCGR achieves the lowest Gini@10 and the highest APLT@10 on all three datasets. Compared with the best-performing baseline on each dataset, PDCGR reduces Gini@10 by 2.91%, 1.27%, and 8.30%, while improving APLT@10 by 5.28%, 3.14%, and 9.09% on ML-100K, ML-1M, and AIV, respectively. In contrast, DiffRec exhibits the highest Gini@10 and the lowest APLT@10, suggesting that diffusion-based preference generation alone does not necessarily alleviate popularity bias. These results demonstrate the complementary roles of the proposed components. Spectral regularization limits the dominance of popularity-related directions during preference recovery, while diffusion-guided interaction completion selects high-confidence relations from unobserved user–item pairs. Together, they promote more balanced item exposure and increase the proportion of long-tail items in recommendation lists.

As shown in Fig. 4, PDCGR achieves the highest Tail-NDCG@10 on all three datasets, outperforming LightGCN by 19.75%, 4.79%, and 13.57%, respectively. Together with the lower AvgPop@10 and Gini@10, this improvement shows that PDCGR reduces head-item dominance while ranking relevant long-tail items more effectively, rather than merely increasing their frequency in recommendation lists.

4.3. PDCGR model analysis

4.3.1. Effectiveness analysis of backbone architectures (RQ3)

To determine the most suitable backbone architecture for PDCGR, we fix the latent-space preference diffusion module (Diff) and design different ranking backbones, covering Diff-MF, Diff-NCF, and Diff-LightGCN, under both settings with and without spectral regularization (SpecReg). Given the recommendation list length K = 10, the corresponding evaluation metrics are denoted as HR@10 and NDCG@10. The recommendation performance of different backbone architectures on the three datasets is shown in Table 5.

As shown in Table 5, without the SpecReg module, the Diff-LightGCN architecture outperforms the Diff-MF and Diff-NCF architectures in HR and NDCG. After incorporating the SpecReg module, all three architectures show clear improvements in HR and NDCG, indicating that the combination of Diff-SpecReg can substantially enhance recommendation performance. Meanwhile, the Diff-SpecReg-LightGCN

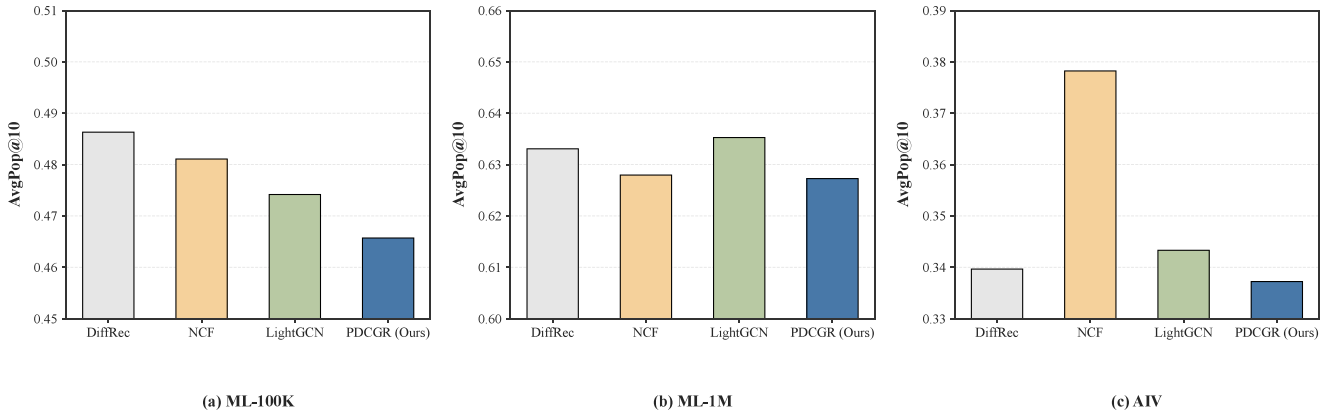


Fig. 2. Popularity bias comparison of different methods (AvgPop@10).

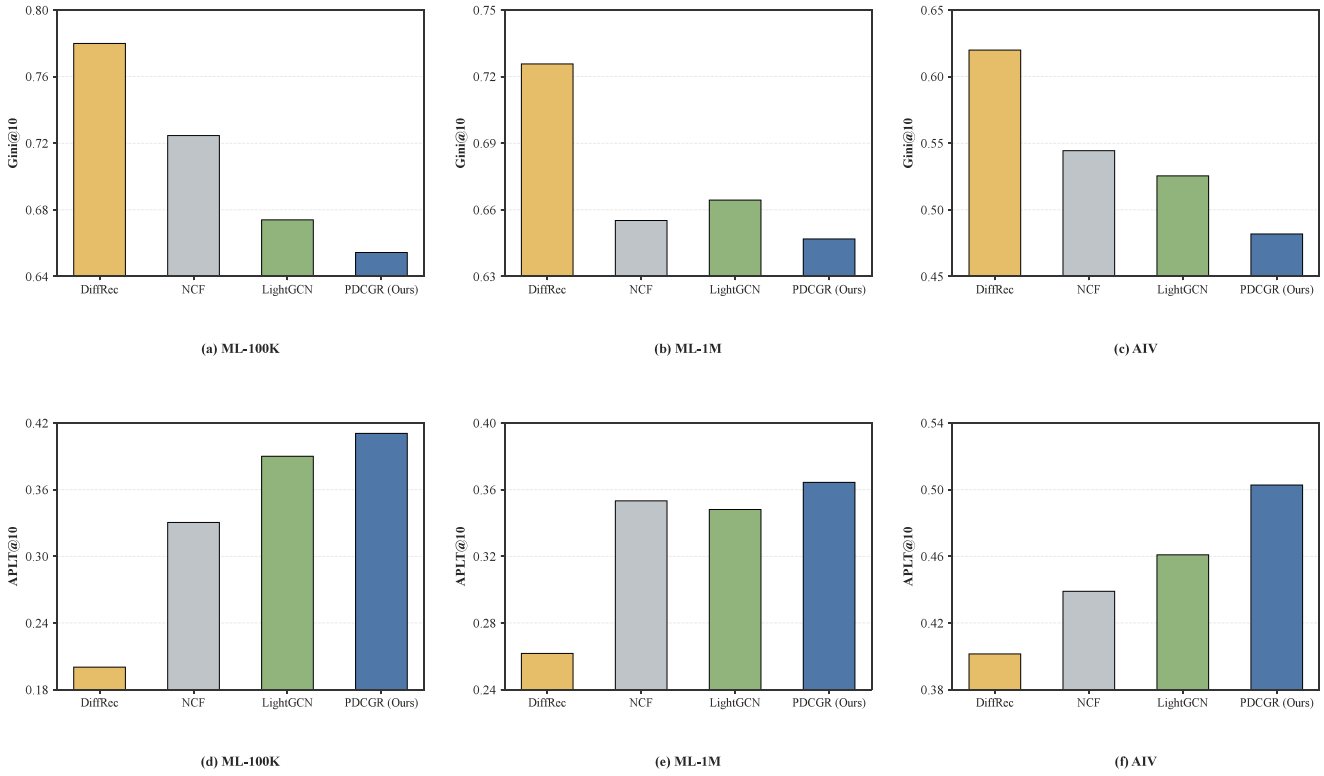


Fig. 3. Item exposure balance and long-tail item proportion comparison (Gini@10 and APLT@10).

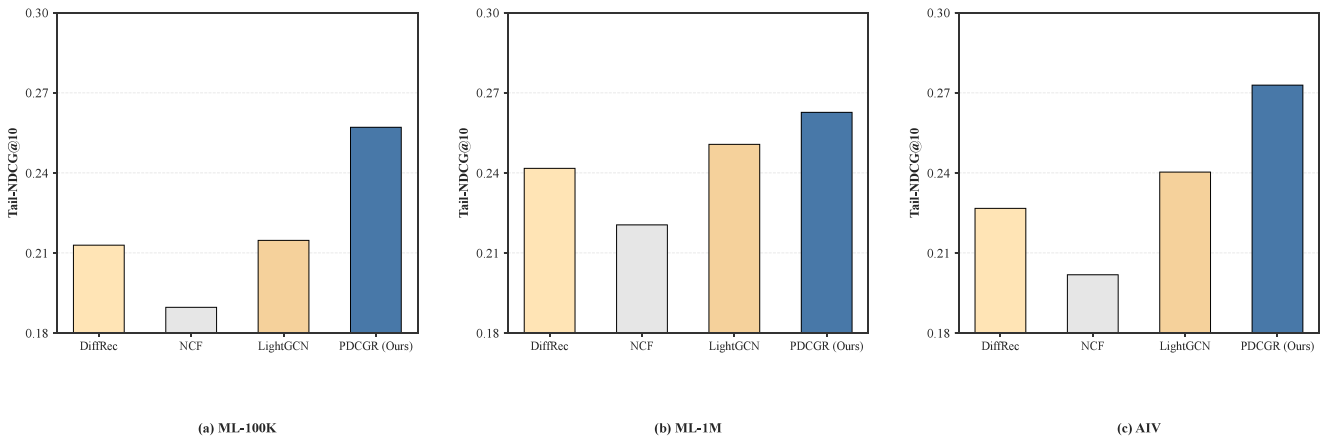


Fig. 4. Long-tail ranking performance comparison (Tail-NDCG@10).

Table 5
Recommendation performance of PDCGR with different backbone models.

Methods	ML-100K		ML-1M		AIV	
	HR@10	NDCG@10	HR@10	NDCG@10	HR@10	NDCG@10
Diff-MF	0.6337	0.4006	0.6379	0.3833	0.5429	0.3459
Diff-SpecReg-MF	0.6690	0.4218	0.6781	0.4052	0.5723	0.3655
Diff-NCF	0.6352	0.3830	0.6545	0.3912	0.5730	0.3683
Diff-SpecReg-NCF	0.6755	0.4113	0.6876	0.4135	0.6135	0.3927
Diff-LightGCN	0.6764	0.4146	0.6907	0.4381	0.6114	0.4081
Diff-SpecReg-LightGCN	**0.7020**	**0.4318**	**0.7312**	**0.4629**	**0.6401**	**0.4238**

architecture outperforms the Diff-SpecReg-MF and Diff-SpecReg-NCF architectures in HR and NDCG. Specifically, compared with the suboptimal Diff-SpecReg-NCF backbone framework, the proposed PDCGR model based on Diff-SpecReg-LightGCN achieves relative improvements of 3.92%, 6.34%, and 4.34% in HR@10 and 4.98%, 11.95%, and 7.92% in NDCG@10 on the ML-100K, ML-1M, and AIV datasets, respectively. These results confirm that incorporating the graph propagation mechanism into Diff-SpecReg-LightGCN enables PDCGR to better capture high-order collaborative information, leading to higher-quality user preference representations and more robust preference vectors.

4.3.2. Impact analysis of different interaction completion strategies (RQ4)

To evaluate the effectiveness and reliability of the interaction completion strategy proposed in Section 3.4, we compare five controlled settings. NoAug uses the original interaction graph without interaction completion. RandomAug randomly selects strictly unobserved user–item pairs. LightGCN-Aug selects high-scoring unobserved pairs according to the dot-product scores obtained from the original LightGCN embeddings. Incl-Train uses diffusion-based scores but retains observed training interactions in the candidate set. PDCGR uses diffusion-recovered latent preference representations to score strictly unobserved user–item pairs. All variants use the same backbone architecture and hyperparameter settings. Except for NoAug, the number of selected edges is fixed at $k_{\text{new}} = 2$. The comparison results on the three datasets are shown in Fig. 5.

As shown in Fig. 5, PDCGR achieves the highest HR@10 and NDCG@10 on all datasets. Compared with NoAug, PDCGR improves HR@10 by 11.57%, 9.10%, and 10.36%, and NDCG@10 by 15.09%, 11.87%, and 11.91% on ML-100K, ML-1M, and AIV, respectively. These results confirm the effectiveness of interaction completion in alleviating data sparsity.

In contrast, RandomAug consistently performs worse than NoAug. On the sparsest AIV dataset, RandomAug decreases HR@10 and NDCG@10 by 5.86% and 8.13%, respectively. This result indicates that random edge addition may introduce noise or potential false positives, especially on highly sparse data. LightGCN-Aug outperforms NoAug but remains inferior to PDCGR, showing that diffusion-recovered latent preference representations provide more effective signals for edge selection than the original LightGCN scores. Incl-Train also performs worse than PDCGR because observed interactions may occupy the limited Top-k_{new} positions, leaving fewer positions for new edges. Taken together, these results demonstrate that the effectiveness of PDCGR depends on both diffusion-guided candidate scoring and the restriction of candidate selection to strictly unobserved interactions, rather than merely adding user–item relations.

4.3.3. Ablation study (RQ5)

To verify the effectiveness of the key components in the proposed PDCGR framework, we conduct systematic ablation experiments by selectively removing different modules and constructing several model variants. The evaluated components include the latent-space preference diffusion module (D), its spectral regularization loss (S), and the interaction completion graph model (ICG). In the ablation study, the full PDCGR model is used as the reference. The three ablation variants are defined as follows:

w/o D: Both the latent-space diffusion module and the spectral regularization term are removed, and LightGCN is trained only on the original user–item graph.

w/o S: The latent-space diffusion module and the interaction completion graph model are retained, while the weight of the spectral regularization loss is set to zero, meaning that no spectral regularization is imposed on the denoised user vectors.

w/o ICG: The latent-space diffusion module and the spectral regularization loss are retained, and the denoised user vectors are directly matched with item vectors through dot-product ranking.

For simplicity, we compare PDCGR with its three ablation variants on ML-100K, ML-1M, and AIV using HR@10 and NDCG@10. The corresponding results are shown in Fig. 6. As shown in Fig. 6, the full PDCGR model consistently outperforms the three ablation variants in terms of HR@10 and NDCG@10 across all datasets. Compared with PDCGR, the w/o D variant reduces HR@10 by 10.37%, 8.34%, and 9.39%, and NDCG@10 by 13.11%, 10.61%, and 10.64% on ML-100K, ML-1M, and AIV, respectively. These results support the role of multi-step diffusion in recovering informative latent preference signals from sparse and potentially noisy implicit feedback, which subsequently provide more reliable information for interaction completion and graph propagation.

Removing the interaction completion graph model also causes clear performance degradation. In particular, NDCG@10 on ML-1M decreases from 0.4629 to 0.3880, while HR@10 on AIV decreases from 0.6401 to 0.5726. This indicates that interaction completion can transform the recovered preference signals into high-confidence user–item relations, enabling graph propagation to capture potential interests and high-order collaborative information more effectively than direct dot-product ranking. When spectral regularization is removed, HR@10 and NDCG@10 on ML-1M decrease by 5.54% and 5.36%, respectively. These results support the effectiveness of spectral regularization in preventing the recovered user representations from becoming excessively concentrated along a few dominant singular directions. Overall, the ablation results demonstrate the complementary contributions of latent preference diffusion, spectral regularization, and interaction completion to the recommendation performance of PDCGR.

4.4. Sensitivity analysis

4.4.1. Effect of diffusion steps

To systematically analyze the effect of the diffusion step number T on recommendation performance, in this experiment, by keeping other hyperparameters unchanged, the HR@10 and NDCG@10 values under different T values are compared on three datasets. The corresponding results are shown in Fig. 7. As shown in Fig. 7, a moderate number of diffusion steps can generate better denoised preference vectors and improve the recommendation performance of PDCGR. When T increases from a small value to a moderate range, the model performance generally improves. This indicates that the multi-step noising and denoising process can gradually reduce the influence of noisy interactions and popularity bias in the latent space, leading to robust user preference vectors. However, when T is further increased, the performance gain gradually saturates, and the benefit brought by additional diffusion steps becomes limited.

4.4.2. Effect of spectral regularization weight

To further explore the effect of the spectral regularization weight β on recommendation performance, in this experiment, by keeping remaining hyperparameters unchanged, β is set to 0, 0.001, 0.01, 0.1, and 0.5, respectively. The resulting changes of HR@10 and NDCG@10 on the three datasets are shown in Fig. 8. Fig. 8 shows that when $\beta = 0$, PDCGR degenerates into a variant that performs diffusion denoising only in the latent space without spectral constraints. In this case, both HR and NDCG are generally lower than those of the settings with spectral regularization. This indicates that diffusion alone is insufficient to effectively suppress the dominant spectral components caused by popular items.

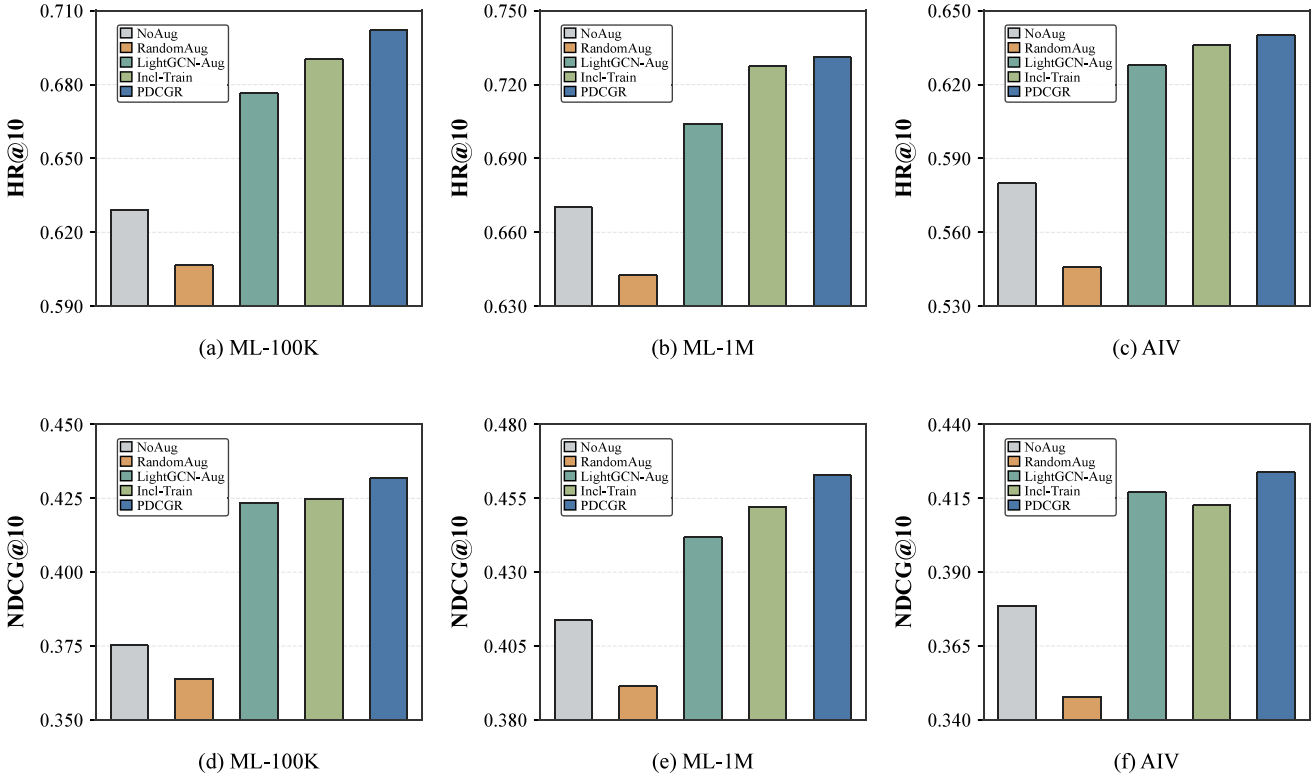


Fig. 5. Performance comparison of different interaction completion strategies (HR@10 and NDCG@10).

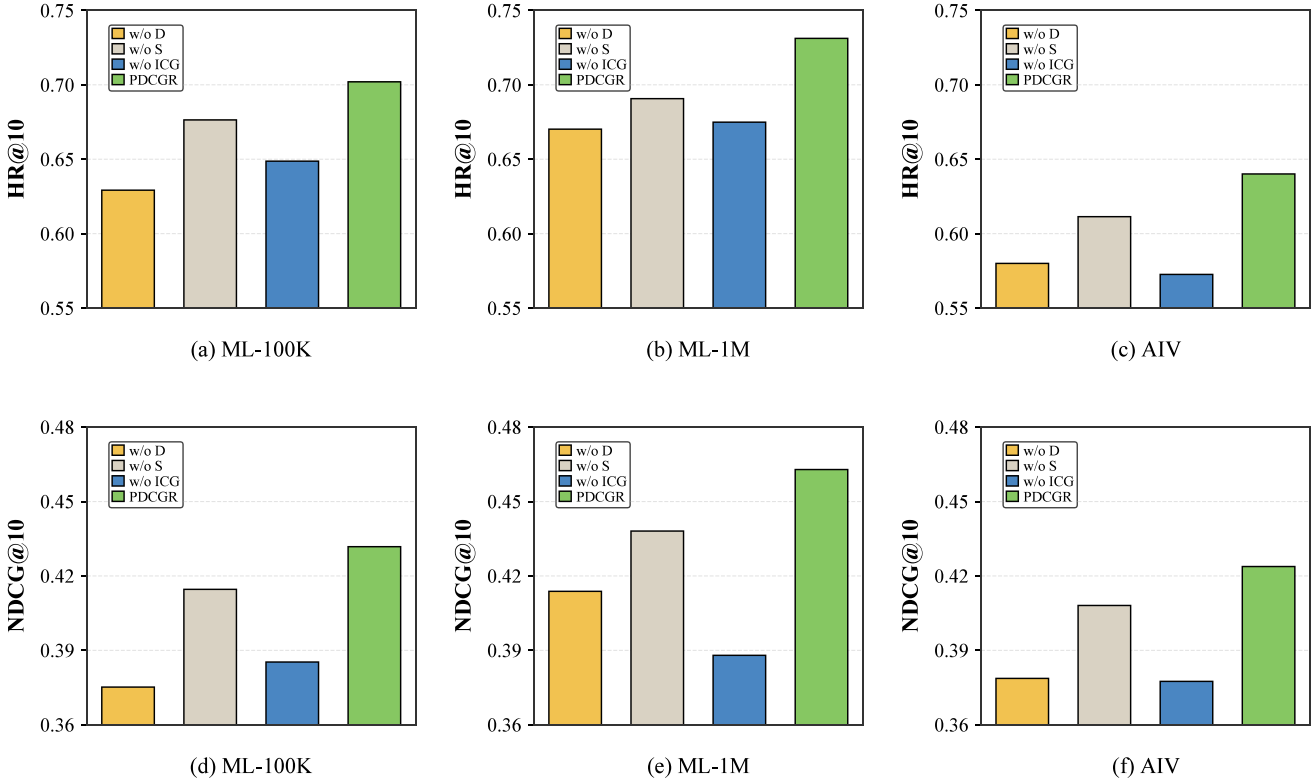


Fig. 6. Ablation study of different sub-modules in PDCGR (HR@10 and NDCG@10).

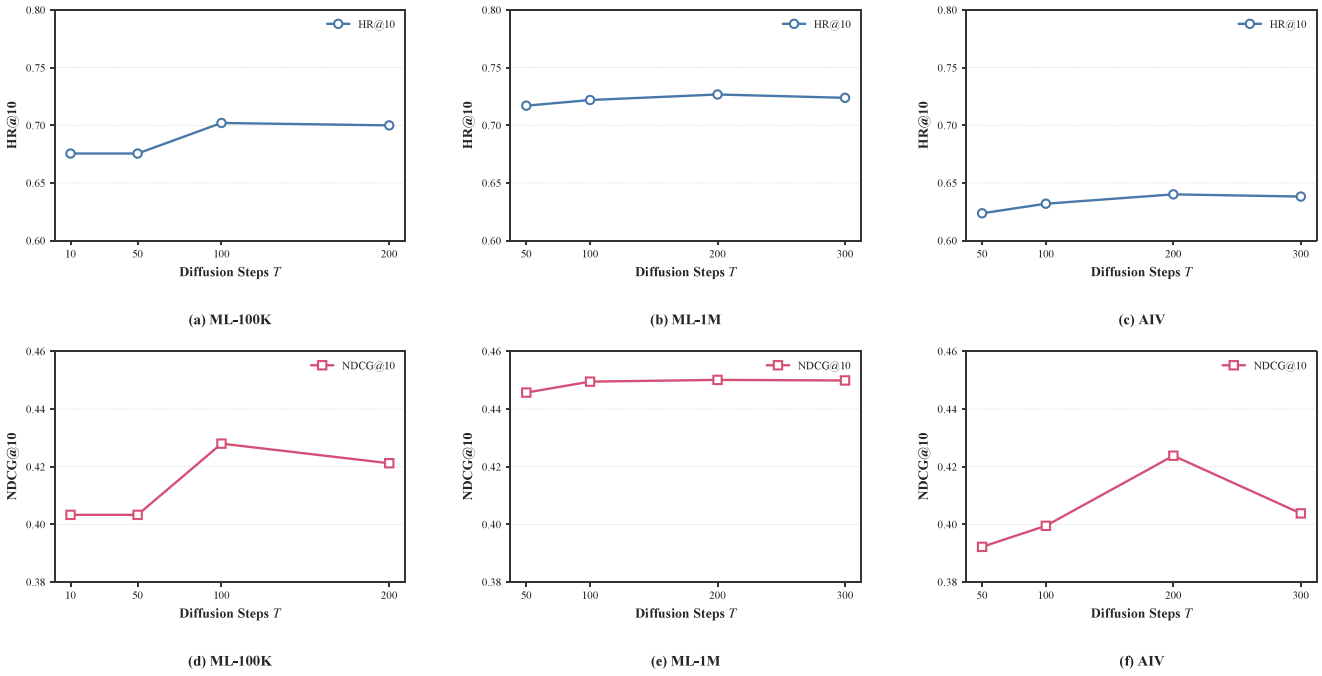


Fig. 7. Effect of diffusion steps T on three datasets.

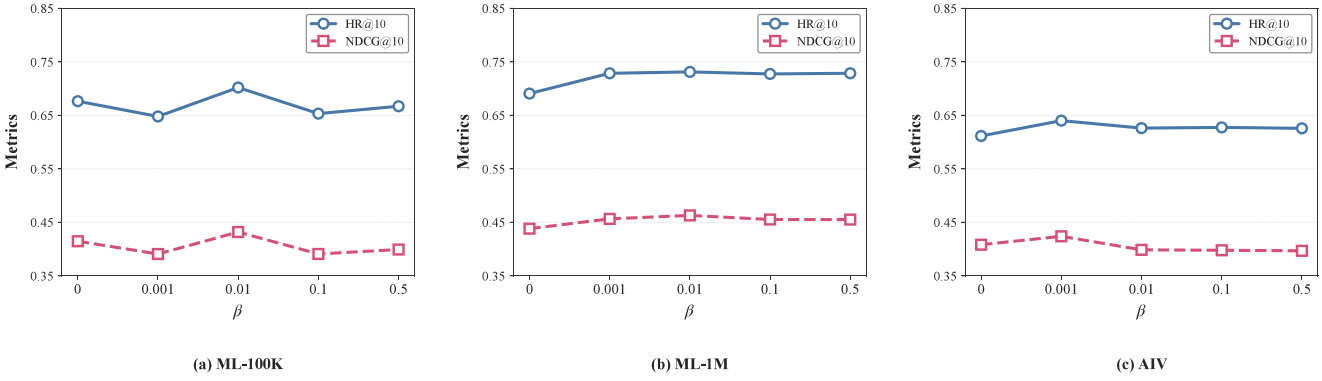


Fig. 8. Effect of spectral regularization weight β on three datasets.

As β gradually increases from 0 to a moderate level, the recommendation performance of the model improves overall. This demonstrates that appropriate spectral regularization can prevent denoised user representations from becoming excessively concentrated along a few dominant singular directions, mitigate popularity bias, and preserve the ability to model long-tail interests. However, when β is further increased to a relatively large value, HR and NDCG on some datasets decline to different degrees. This indicates that overly strong spectral constraints may over-regularize the recovered user representations, which suppresses dominant directions but also weakens personalized differences.

4.4.3. *Effect of interaction completion size*

To systematically analyze the effect of the number of added interactions for each user on the recommendation performance of PDCGR, in this experiment, with all other hyperparameters kept at their default optimal values, k_{new} is set to 0, 1, 2, 3, 5, and 10, respectively. The corresponding HR@10 and NDCG@10 on the three datasets are shown in Fig. 9. As shown in Fig. 9, from the overall trend, when k_{new} increases from 0 to 1 or 2, both HR and NDCG of PDCGR improve steadily on all three datasets. This indicates that an appropriate amount of interaction completion can alleviate the sparsity of the user–item graph and incorporate the high-confidence preferences obtained from diffusion into the graph structure, thereby improving the ranking performance. However, when k_{new} further increases from 3 to 5 or even 10, HR and NDCG begin to level off or even decline, especially on the highly sparse AIV dataset. This suggests that an excessive number of completed interactions may increase the risk of introducing unreliable or potentially false-positive relations, which in turn weakens the ability of graph propagation to exploit the informative preference signals recovered by the spectral-regularized diffusion module.

5. Discussion and implications

The experimental results indicate that the effectiveness of PDCGR under sparse implicit feedback can be attributed to its handling of the trade-off between utilizing unobserved interactions and avoiding the introduction of unreliable signals. RQ1 shows that PDCGR consistently achieves competitive recommendation accuracy across datasets with different scales and sparsity levels. More importantly, RQ4 reveals that the performance gains are contingent upon the reliability of the completed interactions. The diffusion-guided selection strategy, which operates on strictly unobserved user-item pairs, consistently outperforms random selection and a candidate construction approach that retains observed interactions. This suggests that not all unobserved interactions are equally

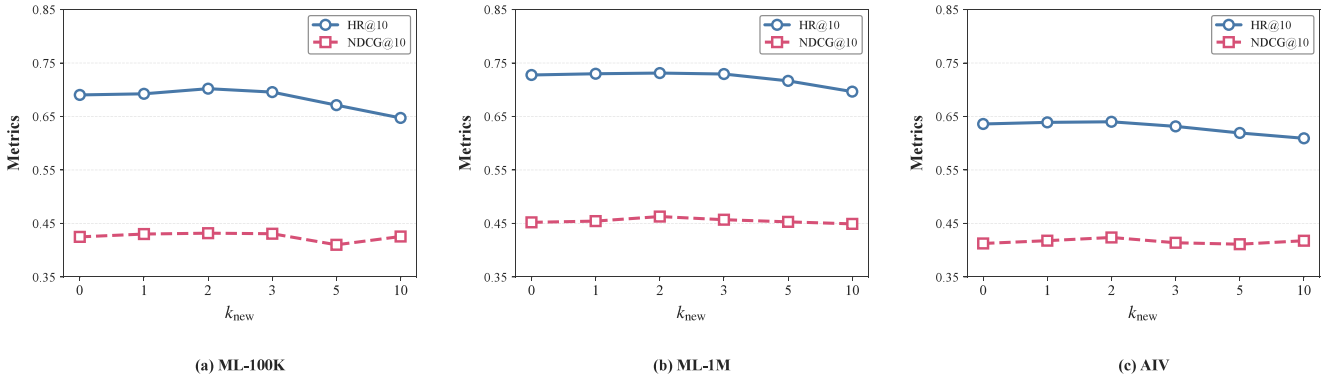


Fig. 9. Effect of interaction completion size k_{new} on three datasets.

useful, and that indiscriminate graph densification may be counterproductive. The analysis of k_{new} further indicates a non-monotonic pattern: adding a modest number of edges per user improves recommendation performance, whereas excessive supplementation tends to introduce unreliable or potentially false-positive relations. RQ5 further clarifies the distinct roles of the proposed components: latent preference diffusion reconstructs informative user preference representations from observed interactions; spectral regularization mitigates the tendency of these representations to become overly concentrated along a few dominant singular directions; and interaction completion leverages the recovered preferences to select high-confidence unobserved interactions for subsequent graph propagation.

The results of RQ2 show that the improvements in recommendation accuracy are accompanied by reduced dependence on popular items, more balanced item exposure, and better long-tail recommendation performance. The observed lower AvgPop@10 and Gini@10 values, together with higher APLT@10 and Tail-NDCG@10 values, indicate that PDCGR increases the presence of long-tail items in recommendation lists and improves their ranking quality. These findings suggest that PDCGR achieves a reasonable balance between recommendation relevance and long-tail exposure. Moreover, the runtime comparison shows that PDCGR requires less per-epoch training time than several diffusion-based baselines, suggesting that its computational cost remains manageable on the evaluated datasets.

Nevertheless, the proposed method still has several limitations. First, PDCGR is designed for static interaction graphs and does not explicitly account for temporal changes in user preferences. Second, it relies primarily on collaborative interaction signals, which limits its applicability to cold-start users and items. Incorporating auxiliary information, such as item text, images, or knowledge graphs, may improve preference recovery in such scenarios. Third, the diffusion training and candidate scoring steps may introduce additional computational overhead when PDCGR is applied to larger item catalogs. Future work may investigate temporal preference modeling, multimodal and knowledge-enhanced conditioning, as well as more efficient diffusion sampling and candidate retrieval strategies, to improve the applicability and scalability of the model.

6. Conclusion

This study proposes a latent Preference Diffusion-based interaction Completion Graph Recommendation model, termed PDCGR, under sparse implicit feedback. PDCGR integrates graph-based representation learning, spectral-regularized latent preference diffusion, and diffusion-guided interaction completion to capture high-order collaborative information, recover informative user preference representations, and select high-confidence interactions from strictly unobserved user–item pairs. Experiments on three public datasets demonstrate that PDCGR consistently improves recommendation accuracy across datasets with different scales and sparsity levels, while producing more balanced item exposure and better long-tail ranking performance. The controlled comparison of interaction completion strategies and the ablation experiments further verify the reliability of the interaction completion strategy and the effectiveness of the proposed components. Overall, PDCGR provides an effective solution for connecting latent preference recovery with interaction completion under sparse implicit feedback.

CRediT authorship contribution statement

Jianmei Ye: Writing – review & editing, Writing – original draft, Validation, Resources, Funding acquisition, Formal analysis, Conceptualization; **Anliang Chen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation; **Jiangzhou Deng:** Writing – review & editing, Validation, Project administration, Methodology, Funding acquisition, Data curation; **Zhiqiang Zhang:** Writing – review & editing, Software, Investigation, Funding acquisition; **Huanyu Chen:** Writing – review & editing, Validation, Investigation; **Yong Wang:** Writing – review & editing, Supervision, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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